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In-orbit thruster calibration techniques and experiment results with UoSAT-12

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Abstract

In this paper, two practical thrusters calibration approaches are proposed, discussed and evaluated through in-orbit flight tests. In the single thruster pulse method, one thruster is commanded to fire for a fixed period to produce a pulsed external torque disturbance. Reaction wheels are used to stabilize the satellite subject to the thruster firings. Magnetorquer coils are switched off during the experiment to reduce the external magnetic disturbances. A preliminary test without thruster firing is used to record the unmodelled external disturbances. The resulting satellite attitude responses and reaction wheel status subject to the thruster firings are recorded for calibration analyses. A batch filter algorithm is developed to estimate the thruster torque coefficients from the recorded experimental data. In the autonomous method, a repetitive torque disturbance (by using a reaction wheel) is exerted to the stabilized satellite, the cold-gas thrusters are then applied in an on-board control system to stabilize the satellite attitude. By recording the known attitude disturbances and the thruster controller outputs, a recursive least-squares algorithm is used to process the recorded data and to estimate the thruster parameters. Both simulation and in-orbit test results are presented to evaluate the performance and design objectives. The in-orbit test results converge to the same values using the two different approaches.
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Keywords: Thruster calibration; Satellite; Parameter estimation; In-orbit experiment

1. Introduction

Many space missions, such as high-resolution imaging, require a high accuracy in attitude determination and control system (ADCS). Thrusters or gas jets are one of the common actuators on-board spacecraft for attitude and orbit control. To perform a precise manoeuvre, accurate knowledge about the thrust level and thrust direction become essential. In-orbit calibration of attitude control hardware, such as thrusters, is often necessary in practical mission operations (Wiktor, 1996). Thruster calibration is also important to accurately estimate and preserve fuel to maximize the spacecraft's life (Prickett & Hoang, 1992).

Accurate calibration of the thrusters prior to flight is limited by various factors. Among these is the difficulty of maintaining a suitable vacuum in a chamber during thruster calibration, while gas is discharged from the thruster. Commonly, it is difficult to predict and establish during ground testing the exact temperature and pressure conditions under which the thrusters will operate while in orbit. The output direction of the thrust will not necessarily be concentric with the nozzle. To measure the true direction and magnitude parameters of the thruster force are very difficult on the ground, especially considering plume impingement (Parvez, 1990), which will further affect the direction of a thruster force output. If these parameters are not properly identified in orbit, significant attitude and orbit control errors may result.

Modern estimation theory and system identification techniques provide the possibility to implement on-board calibration procedures for these actuators. Although this will increase the software complexity of the on-board computer, it will provide better performance knowledge after launch in order to improve the future design of accurate attitude and orbit controllers. This paper presents the principles and performances of

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two practical methods to calibrate thrusters on-board satellites. One method uses a single firing pulse to disturb the satellite's attitude, the data is then batch processed to estimate the thruster parameters. The other method estimates the thruster parameters recursively by repetitive firings and can be used in an autonomous on-board application.

UoSAT-12 is a low-cost 320 kg minisatellite built by Surrey Satellite Technology Ltd. It is a technology demonstrator for high performance attitude control and orbit maintenance for a future constellation of earth observation satellites. The satellite uses a three-axis reaction wheel and a cold gas thruster system to enable precise and fast control of its attitude. Magnetorquer coils assist the wheels mainly for momentum dumping. Ten cold-gas thrusters can be used in various combinations for both attitude and orbit control and a single N_2O resisto-jet is used exclusively for orbit maintenance. The architecture of the cold-gas thruster system is illustrated in Fig. 1, where three big tanks are used to store the original high-pressure Nitrogen gas at 200 bar, with a capacity of 27 l. Before reaching to the thruster throttles, the high-pressure gas will first be released into an accumulator tank where the pressure is controlled to be at the certain level of 4 bar. Then a set of valve logics is used to switch on and off the 10 cold thrusters. As the gas reaching to thrusters is controlled constantly at the level of 4 bar in the accumulator, the resulting thrusters level is then not affected by the real pressure in the tanks, which changes along with the usage of gas.

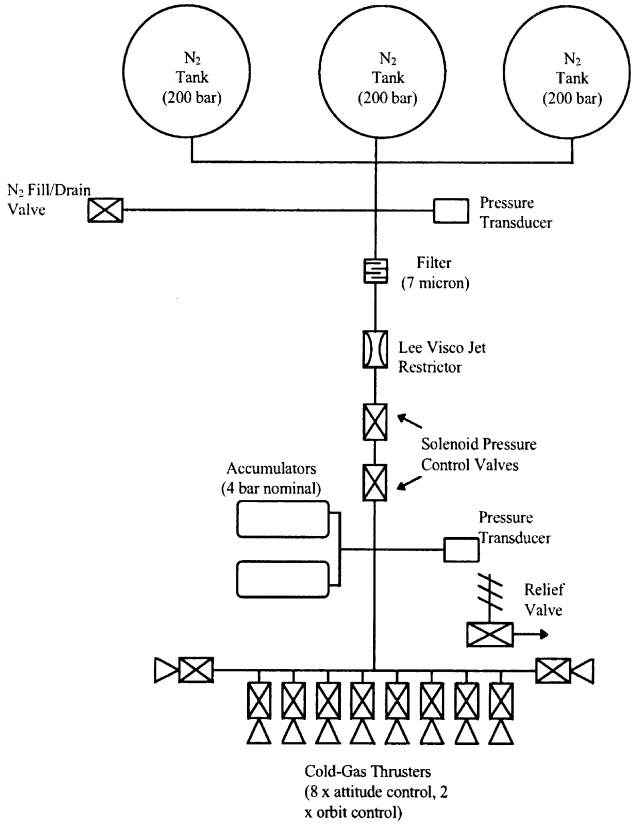


Fig. 1. Cold-gas thrusters system on-board UoSAT-12.

This paper will discuss the in-orbit thruster calibration techniques based on this cold-gas thrusters system on-board UoSAT-12. It is organized as follows: Section 2 introduces the dynamic model relations and basic attitude controllers on-board the satellite. Section 3 discusses the single thruster pulse method. Section 4 presents the principle of the autonomous method. Simulation test results are presented in Section 5, while in-orbit test results on-board UoSAT-12 are discussed in Section 6. The final section summarizes the research results and presents some conclusions.

2. Spacecraft dynamics and attitude controller

The dynamic behavior of a low Earth orbit (LEO) Earth-pointing satellite, using three-axis reaction wheels as internal torque actuators and magnetorquers and thrusters as external torque actuators, is described by the following equation (Steyn, 1995):

$$\mathbf{I}\dot{\boldsymbol{\omega}}_B^I = \mathbf{N}_T + \mathbf{N}_M + \mathbf{N}_{GG} + \mathbf{N}_D - \boldsymbol{\omega}_B^I \times (\mathbf{I}\boldsymbol{\omega}_B^I + \mathbf{h}) - \dot{\mathbf{h}}, \quad (1)$$

where $\boldsymbol{\omega}_B^I$, $\mathbf{I}$, $\mathbf{h}$, $\mathbf{N}_M$, $\mathbf{N}_T$, $\dot{\mathbf{h}}$, $\mathbf{N}_D$ and $\mathbf{N}_{GG}$ are respectively the inertially referenced body angular velocity vector, moment of inertia of spacecraft, three-axis reaction wheel angular momentum vector, applied torque vector by three-axis magnetorquers, applied torque vector by three-axis thrusters, applied torque vector by three-axis reaction wheels, external disturbance torque vector including the torques due to the aerodynamic and solar forces, and gravity gradient torque vector.

The attitude of a satellite can be presented by means of a quaternion (Wertz, 1989). The use of a quaternion in describing the orientation of a rigid body lends itself well to on-board calculation since no trigonometric relations or singularities (which arise when using Euler angles) exist in the kinematic formula, and it can easily be transferred to the orbit-referenced coordinate system. The parameterization of the quaternion vector is

$$q_1 \equiv e_{xo} \sin\left(\frac{\Phi}{2}\right), \quad q_2 \equiv e_{yo} \sin\left(\frac{\Phi}{2}\right),$$
$$q_3 \equiv e_{zo} \sin\left(\frac{\Phi}{2}\right), \quad q_4 \equiv \cos\left(\frac{\Phi}{2}\right), \quad (2)$$

where e_{xo}, e_{yo}, e_{zo} are the components of the Euler vector in orbit-referenced coordinates and Φ is the rotation angle around the Euler vector. The kinematics can then be updated by the following vector set of differential equations:

$$\dot{\mathbf{q}} = \frac{1}{2}\boldsymbol{\Omega}\mathbf{q}, \quad (3)$$

$$\mathbf{\Omega} = \begin{bmatrix} 0 & \omega_{oz} & -\omega_{oy} & \omega_{ox} \\ -\omega_{oz} & 0 & \omega_{ox} & \omega_{oy} \\ \omega_{oy} & -\omega_{ox} & 0 & \omega_{oz} \\ -\omega_{ox} & -\omega_{oy} & -\omega_{oz} & 0 \end{bmatrix}, \quad (4)$$

where $\boldsymbol{\omega}_B^O = \begin{bmatrix} \omega_{ox} & \omega_{oy} & \omega_{oz} \end{bmatrix}^{\mathrm{T}}$ is the orbit-referenced angular body rate vector. The relation between $\boldsymbol{\omega}_B^o$ and $\boldsymbol{\omega}_B^I$ for a circular orbit is given by,

$$\boldsymbol{\omega}_B^I = \boldsymbol{\omega}_B^O + A \cdot \begin{bmatrix} \mathbf{0} & -\omega_o & \mathbf{0} \end{bmatrix}^{\mathrm{T}}, \quad (5)$$

where A is the attitude transformation matrix from orbital coordinates to body fixed coordinates, ω_o is the mean orbital angular rate, which is considered to be a constant value, computed by $\omega_o = \mathbf{2\pi}/T_o$, where T_o is the average orbital period. To comply with the on-board Kalman filter for attitude estimation, the Euler 213 transformation matrix A_{213} (Sidi, 1997) is applied in this study.

Whenever quaternions are used directly in the attitude control laws, it is convenient to define the error quaternion as the difference between the current quaternion and the commanded quaternion. It can be represented by

$$\mathbf{q}_e = \begin{bmatrix} q_{1e} \\ q_{2e} \\ q_{3e} \\ q_{4e} \end{bmatrix} = \begin{bmatrix} q_{4c} & q_{3c} & -q_{2c} & -q_{1c} \\ -q_{3c} & q_{4c} & q_{1c} & -q_{2c} \\ q_{2c} & -q_{1c} & q_{4c} & -q_{3c} \\ q_{1c} & q_{2c} & q_{3c} & q_{4c} \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix}, \quad (6)$$

where q_{ie} are the components of the error quaternion, q_{ic} the components of the commanded quaternion and q_i the components of the current orbit-referenced spacecraft body quaternion.

For three-axes stabilization and pointing control, a modified version of the globally stable quaternion PD feedback control law (Wie, Weiss, & Arapostathis, 1989) was implemented on UoSAT-12. It consists of linear error-quaternion feedback, with linear and non-linear body-rate feedback terms to compensate for the gyroscopic coupling torques. The control torque vector is represented as

$$\mathbf{N}_{W/T} = \boldsymbol{\omega}_B^I \times (\mathbf{I}\boldsymbol{\omega}_B^I + \mathbf{h}) - k_p \boldsymbol{\omega}_B^O - k_d \mathbf{I}\mathbf{q}_{vec}, \quad (7)$$

where k_p, k_d are positive gain scalars, $\mathbf{N}_{W/T}$ is the control torque vector implemented by three-axis reaction wheels or thrusters, and $\mathbf{q}_{vec}$ is the vector part of error quaternion.

The controller gains k_p, k_d are chosen to satisfy certain requirements for the closed-loop system performance, see Wie et al., 1989 for details.

3. Single thrust pulse method

As the satellite orbits, any externally applied torques or disturbances will induce immediate changes to the satellite attitude and long-term changes to the orbit parameters. In the single thruster pulse method, a thruster is fired for a fixed time period, i.e., a thruster pulse is exerted to the satellite. The attitude and internal torque changes are recorded (sampled) and analyzed (batch processed) to estimate the magnitude of the thruster-related control torques.

Suppose that the satellite is stabilized using a PD type quaternion feedback controller (7) with 3-axis reaction wheels. When a thruster is fired for a short time period ΔT, its effects on the satellite dynamic behavior would be reflected by changes to the satellite attitude and internal torque actuators. Integrating Eq. (1) yields,

$$\int_{t_0}^{t} \mathbf{I}\dot{\omega}_B^I \, \mathrm{d}t = \mathbf{N}_T \Delta T + \int_{t_0}^{t} (\mathbf{N}_M + \mathbf{N}_{GG} + \mathbf{N}_D) \, \mathrm{d}t - \int_{t_0}^{t} (\omega_B^I \times (\mathbf{I}\omega_B^I + \mathbf{h})) \, \mathrm{d}t - \int_{t_0}^{t} \dot{\mathbf{h}} \, \mathrm{d}t \quad (8)$$

and define

$$\mathbf{S} = \mathbf{N}_T \Delta T + \int_{t_0}^{t} (\mathbf{N}_M + \mathbf{N}_{GG} + \mathbf{N}_D) \, \mathrm{d}t = \mathbf{I}\Delta\omega_B^I + \int_{t_0}^{t} (\omega_B^I \times (\mathbf{I}\omega_B^I + \mathbf{h})) \, \mathrm{d}t + \mathbf{h}(t) - \mathbf{h}(t_0). \quad (9)$$

The purpose of thruster calibration is to determine the thruster-induced torque $\mathbf{N}_T$ as accurately as possible. In Eq. (9), the time period ΔT can be assigned accurately during the firing test. If the other terms in the equation could be determined, then the value of $\mathbf{N}_T$ can be calculated accurately. This is the basic idea of the single thruster pulse method.

From the on-board attitude sensors, i.e. the horizon sensor, magnetometer, and sun sensors, the attitude information of the satellite could be measured and recorded. In practice, an on-board Kalman filter could provide attitude and angular rate estimation to a good accuracy. Furthermore, the speed of each reaction wheel can be measured to a high accuracy. As the moment of inertia of each wheel is also known accurately, the reaction wheel's angular momentum vector can then be calculated. Thus, the right side of Eq. (9) could then be calculated from the in-orbit recorded flight information.

Looking at the left side of Eq. (9), of the three disturbance terms, the magnetic torque uncertainty can have the biggest contribution and furthermore, its value and direction depend on the local geomagnetic field. To reduce its influence, the magnetorquer controller on-board the satellite is switched off during the firing tests. Furthermore, to compensate for the influences of the three uncertain torque terms, an equivalent test, where no thruster is fired and the magnetorquer controller is switched off, is conducted proceeding the thruster test The corresponding information are then used to

estimate these influences, that is,

$$\mathbf{S}_0 = \int_{t_0}^{t} (\mathbf{N}_M + \mathbf{N}_{GG} + \mathbf{N}_D)\,\mathrm{d}t = \mathbf{I}\Delta\omega_B^I + \int_{t_0}^{t} (\omega_B^I \times (\mathbf{I}\omega_B^I + \mathbf{h}))\,\mathrm{d}t + \mathbf{h}(t) - \mathbf{h}(t_0). \quad (10)$$

Then the thruster parameters are estimated, i.e., calibrated, using the following algorithm:

$$\mathbf{N_T} = \frac{\mathbf{S} - \mathbf{S_0}}{\Delta T}. \quad (11)$$

With this method, the in-orbit test scheme can be implemented in two stages: The first stage is called the preparation stage, aiming to record all the in-orbit external disturbances. From steady flight conditions, with the magnetorquers disabled, the flight state information is recorded for a certain time period. The second stage, called the firing test stage, records the overall external disturbances with a single thrust pulse. During this stage, the magnetorquers are still disabled, one thruster is commanded to fire for a short period, and the flight state changes and other related flight information are recorded for a certain time period. During the complete test, a PD type quaternion feedback controller with reaction wheels is applied to stabilize the satellite.

The flight state information to be recorded during the tests include: (1) three-axis satellite attitude information, i.e., the roll, pitch and yaw angles; (2) the speed of each reaction wheel in operation. Batch algorithms are then used to estimate the thruster-induced torque parameters.

Firstly, with no gyro rate measurements available, the body angular rate can be estimated from the recorded flight attitude information. For simplicity, the following one-step differential algorithm can be used (for small attitude angles),

$$\hat{\boldsymbol{\omega}}_B^O(i) = \frac{\mathbf{RPY}(i) - \mathbf{RPY}(i-1)}{T_s}, \quad (12)$$

$$\hat{\boldsymbol{\omega}}_B^O(i) = \hat{\boldsymbol{\omega}}_B^O(i) + A(i) \cdot \begin{bmatrix} \mathbf{0} & -\omega_0 & \mathbf{0} \end{bmatrix}^{\mathrm{T}}, \quad (13)$$

where $\mathbf{RPY}(i)$ is the *ith* recorded flight attitude vector (roll, pitch and yaw small attitude angles), $\hat{\boldsymbol{\omega}}_B^O(i)$ is the estimated body angular rate referenced to orbit coordinates at the *ith* sampling time, $\hat{\boldsymbol{\omega}}_B^O(i)$ is the estimated inertial body angular rate vector, and $\mathbf{A}(i)$ is the attitude transformation matrix from orbital coordinates to body fixed coordinates.

The angular momentum of the reaction wheels are simply computed by multiplying their inertial tensor with their corresponding rotation rate vector,

$$\mathbf{h} = \mathbf{I_{RW}} \cdot \boldsymbol{\omega}_{\mathbf{RW}} = \begin{bmatrix} I_{RWx} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & I_{RWy} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & I_{RWz} \end{bmatrix} \cdot \begin{bmatrix} \omega_{RWx} \\ \omega_{RWy} \\ \omega_{RWz} \end{bmatrix}, \quad (14)$$

where $\mathbf{I_{RW}}$ is the inertial tensor of the three-axis reaction wheel system, $\boldsymbol{\omega}_{\mathbf{RW}}$ is the angular rate vector of the three reaction wheels.

With the obtained values for $\hat{\boldsymbol{\omega}}_{\mathbf{B}}^{\mathbf{I}}(i)$ and $\mathbf{h}$, numerical integration is then required to estimate the thruster parameters according to Eq. (9). For example the following Trapezoidal method can be applied for numerical integration,

$$y_{i+1} = y_i + \frac{T_s}{2}[f(\mathbf{x}_i) + f(\mathbf{x}_{i+1})], \quad (15)$$

where f is the functions to be integrated, T_s is the sampling period of recording flight information. Here $\mathbf{x}$ actually represents the time varying state vector.

With this test scheme, the calibration accuracy of the thruster parameters are affected directly by the following factors.

- *Attitude measurement accuracy*. The estimation is based on these measurements from the on-board attitude sensors and filters. To improve this accuracy, the sun sensors on-board UoSAT-12 have been calibrated with in-orbit flight results (Wu & Steyn, 2002).
- *Accuracy of the satellite moment of inertial tensor (MOI)*. From Eq. (9), the accuracy in MOI gives direct influences on the calibration accuracy.
- *Sample time to record the flight state information*. This time interval has a direct influence on the computation of the body angular rate, by Eq. (12). In principle, the smaller this sample time, the better the accuracy of the calibration. However, due to the limit of the on-board computer's (OBC) resources, the time intervals to record all the required flight information are limited.
- *Accuracy of the angular momentum measurement of the reaction wheel system*. The angular momentum of the reaction wheel system is directly included in the computation. Thus its accuracy exerts direct influences onto the calibration results.
- *Time variation of the disturbance torques*. Although two test stages are designed to compensate for the influences of external torque disturbances, there are still errors due to the time difference between the two stages. Care should be taken to reduce these errors as much as possible.
- *The firing time length of thruster pulse*. In principle, a longer firing will cause a bigger disturbance to the satellite attitude, increase the change in angular momentum, and thus decrease the effect of uncertain external disturbances. However, when the attitude change exceeds the sensor measurement range, the attitude measurement accuracy would be decreased. A criteria to determine the thruster firing period is to guarantee that the maximum attitude disturbance should stay within the sensors FOV, and the reaction saturation values are not exceeded. Simulation tests

can be used to determine an appropriate firing time duration for the in-orbit tests of each thruster.

4. Autonomous method

Following an idea of Wiktor (1996), a novel recursive least-squares (RLS) algorithm was developed to calibrate the cold gas thrusters autonomously (real-time and on-line) during normal mission conditions. In this method, a periodic disturbance torque is produced using the reaction wheels, while the attitude is controlled using the thrusters through the control law of Eq. (7).

For the applied disturbance torque, a Bang–Bang controller is implemented for the reaction wheels. This controller is based on a linear PD feedback method and a two level non-linear switching control law. The switching function is determined from a linear feedback equation and a hysteresis band, as expressed by the following equation:

$$\mathbf{e} = \mathbf{K}_1\mathbf{q}_{vec} + \mathbf{K}_2\boldsymbol{\omega}_B^O,$$

$$\mathbf{N}_W = \begin{cases} -\mathbf{N}_{max} & \text{for } \mathbf{e} \geqslant \mathbf{e}_{band}, \quad \text{with } \mathbf{e} \text{ increasing,} \\ \mathbf{N}_{max} & \text{for } \mathbf{e} \leqslant -\mathbf{e}_{band}, \quad \text{with } \mathbf{e} \text{ decreasing,} \end{cases} \tag{16}$$

where $\mathbf{K}_1$ and $\mathbf{K}_2$ are controller gains, $\mathbf{e}$ is the control error and $\mathbf{e}_{band}$ is the error hysteresis band. This error band is tuned to adjust the level of the wheel momentum build-up indirectly (and the error quaternion). When reducing the error band, the torque output switches at a higher frequency and the wheel momentum build-up is reduced. An increase in the error band has the opposite effect. The controller gains $\mathbf{K}_1$ and $\mathbf{K}_2$ adjusts the slope of the switching function, this also controls the stability of the non-linear limit cycle and the magnitude of the error quaternion. The controller gains $\mathbf{K}_1 = k_1\mathbf{I}$ and $\mathbf{K}_2 = k_2\mathbf{I}$ are optimised during simulation by using trial and error method.

The Bang–Bang controller switches between two control torques (positive and negative). Fig. 2 explains how the Bang–Bang controller works. Suppose we have a certain attitude and rate at point $t_o(\phi, \dot{\phi})$, a positive wheel torque (negative satellite torque) is used to enable a trajectory from point t_o until the switching line (point t_1) is reached, then the controller switches to a negative wheel torque to enable a trajectory from point t_1 until the other switching line (point t_2) is reached. The controller switches again. Finally, it alternates continuously between points t_3 and t_4 on the switching lines within the error band, and thus produces a periodic disturbance torque to the satellite.

Suppose the n thrusters are controlled by n commands in vector $\mathbf{T}_c$ from the attitude controller, the resultant torque $\mathbf{N}_T$ exerted on the spacecraft from this set of thruster commands is

$$\mathbf{N}_T = \mathbf{A}_c\mathbf{T}_c \tag{17}$$

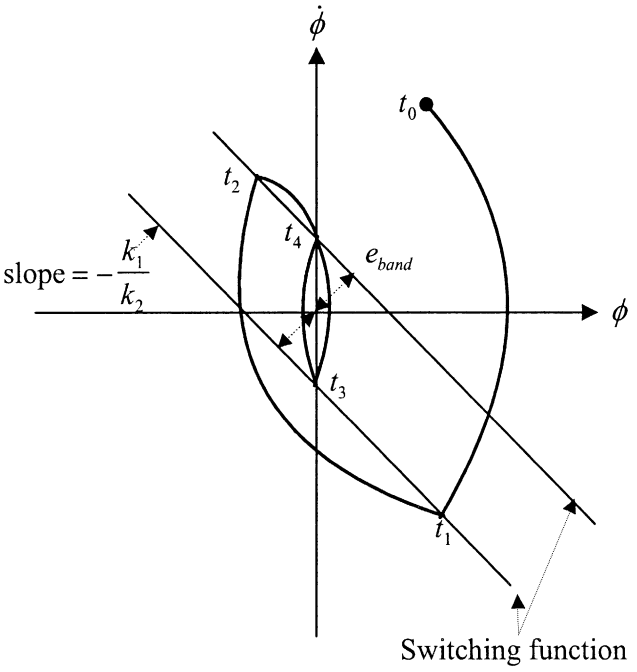


Fig. 2. Phase plane path of a Bang–Bang control system with linear PD feedback.

where $\mathbf{N}_T$ is a 3×1 vector, and $\mathbf{A}_c$ the $3 \times n$ thruster configuration matrix which contains the information about the output direction and magnitude of each thruster. The goal of the thruster calibration is to identify these $3 \times n$ coefficients in spite of sensor noise and external disturbance torques.

The calibration scheme is illustrated in Fig. 3. A known disturbance torque $\mathbf{N}_W(k)$ is applied to the spacecraft using reaction wheels controlled by the Bang–Bang control law (16), whilst the attitude controller (7) sends commands $\mathbf{T}_c(k)$ to the thrusters (generating torques $\mathbf{N}_T(k)$) to compensate for both this known and also any unknown external disturbance torques $\mathbf{N}_D(k)$.

The in-orbit thruster calibration procedure (below the dotted line in Fig. 3) uses the known disturbance torque $\mathbf{N}_W(k)$ and the resulting measured satellite state to calculate the calibration torque $\mathbf{N}_c(k)$ acting on the satellite. This calibration torque is equal to the sum of the known and the unknown disturbance torques and can be obtained by rearranging Eq. (1) thus

$$\begin{aligned} \mathbf{N}_c &= \mathbf{I}\dot{\boldsymbol{\omega}}_B^I + \boldsymbol{\omega}_B^I \times (\mathbf{I}\boldsymbol{\omega}_B^I + \mathbf{h}) - \mathbf{N}_W \\ &= \mathbf{N}_T + \mathbf{N}_D + \mathbf{N}_{GG} + \mathbf{N}_M, \end{aligned} \tag{18}$$

where $\mathbf{N}_c = \begin{bmatrix} N_{cx} & N_{cy} & N_{cz} \end{bmatrix}^{\mathrm{T}}$ is the calibration torque vector, $\mathbf{N}_W = -\mathbf{h}$ is the reaction wheel torque vector, produced by the Bang–Bang controller. In the computation of Eq. (18), the same algorithms of Eqs. (12)–(14) are applied to process the recorded flight attitude information.

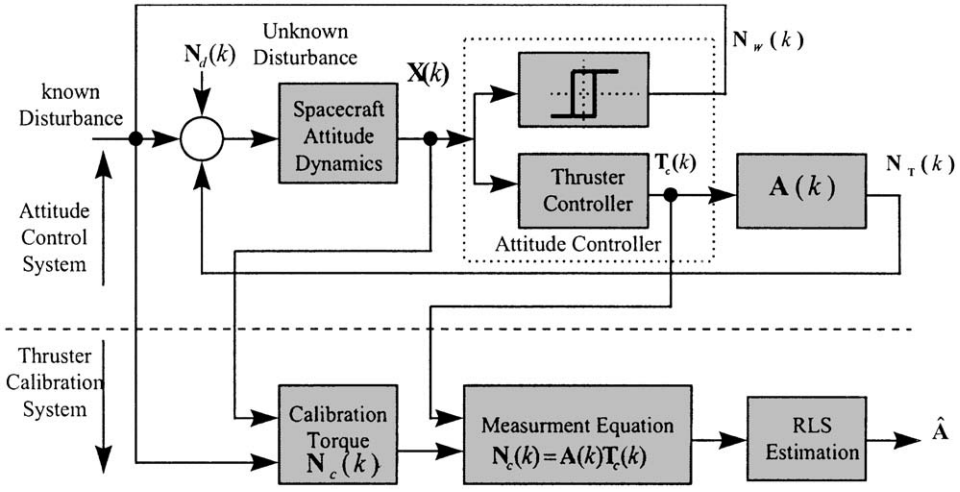


Fig. 3. Autonomous thruster calibration block diagram.

If the satellite attitude is maintained and other disturbances ($\mathbf{N}_D$, $\mathbf{N}_{GG}$, and $\mathbf{N}_M$) neglected, this known disturbance torque is then exactly balanced by the thruster torque given by Eq. (17). The above relation can be rewritten as a linear regression problem (Åström & Wittenmark, 1989):

$$\mathbf{y} = \mathbf{N}_c^{\mathrm{T}} = \mathbf{T}_c^{\mathrm{T}} \cdot \hat{\mathbf{A}}_c^{\mathrm{T}} + \mathbf{v}, \tag{19}$$

where $\mathbf{v}$ represents the effect of measurement noise plus the unknown external disturbance torques, and $\hat{\mathbf{A}}_c$ is the thruster parameters to be estimated.

To estimate the model parameters in real-time, a RLS calibration algorithm was developed for improved convergence and accuracy. The algorithm is a recursive implementation of the least-squares minimization technique and is appropriate because of the almost time invariant nature of the thrusters parameters $\hat{\mathbf{A}}_c$. The error to be minimized can be written as a standard least-squares parameter estimation problem for the linear regression model Eq. (19):

$$\mathbf{e}(k) = \mathbf{y}(k) - \Phi(k)\boldsymbol{\theta}(k), \tag{20}$$

where $\mathbf{y}(k)$ is the observed variable, $\boldsymbol{\theta}(k) = \hat{\mathbf{A}}_c^{\mathrm{T}}$ is the unknown parameters to be estimated, and $\Phi(k) = \mathbf{T}_c^{\mathrm{T}}$ is the known functions. The least-squares cost function to be minimized is taken as:

$$J = \frac{1}{2}\sum \lambda^{t-k}\mathbf{e}^{\mathrm{T}}(k)\mathbf{e}(k). \tag{21}$$

The forgetting factor λ is a constant ($\lambda \leqslant \mathbf{1}$) that defines the smoothing of the estimates by introducing a time varying weighting of data. The full RLS algorithm is summarized as follows (Åström & Wittenmark, 1989; Bordany, 2001):

- Compute the regression matrix $\Phi(k)$ and the residual $\mathbf{e}(k)$ from (20).
- Compute the update gain vector

$$\mathbf{K}(k) = \mathbf{P}(k-1)\Phi(k)[\lambda\mathbf{I} + \Phi^{\mathrm{T}}(k)\mathbf{P}(k-1)\Phi(k)]^{-1}, \tag{22}$$

  where, $\mathbf{P}$ is defined as the covariance matrix of the parameter vector $\theta(k)$. The initial value $\mathbf{P}(0)$ determines the initial convergence speed. During this experiment, it was chosen to give the optimal convergence response during simulation (no overshoot or a too slow exponential response).
- Update the parameter vector

$$\boldsymbol{\theta}(k) = \boldsymbol{\theta}(k-1) + \mathbf{K}(k)\mathbf{e}(k). \tag{23}$$

- Update the covariance matrix

$$\mathbf{P}(k) = [\mathbf{I} - \mathbf{K}(k)\Phi^{\mathrm{T}}(k)]\mathbf{P}(k-1)/\lambda. \tag{24}$$

The above general RLS procedure is further modified by processing the error to remove low frequency components and outlying values. To improve the robustness of the RLS algorithm, the error can be modified by a non-linear saturation function (Steyn, 1995) as follows:

$$f\{e_i(k)\} = \frac{e_i(k)}{\mathbf{1} + b|e_i(k)|} \quad (i = 1, 2, \ldots, n). \tag{25}$$

The constant b is defined such that the function is still linear for normal values of $e_i(k)$, while decreasing the influence of large outliers.

5. Simulation studies

Digital simulation tests were extensively used to validate and compare the thruster calibration approaches, to evaluate their performances, and to help determine appropriate in-orbit experiment parameters. Simulations were performed in the Matlab/Simulink environment. The UoSAT-12 satellite in a LEO was used as the test bed, where the satellite dynamics and

kinematics were modeled and implemented in Matlab/Simulink. The parameters used during the simulation are chosen the same as in the practical satellite, as summarized in Table 1.

For the single thruster pulse method, a series of simulation tests were conducted to investigate the influence of various parameters on the calibration accuracy. The simulation results and some findings are summarized next, while detailed simulation results can be found in Steyn, Wu, and Bordany (2001).

- *Simulation time step*. To keep a satisfactory accuracy, the simulation time step must be less than the firing time period, and should be chosen such that the firing time period becomes a multiple of the time step. As the firing time is normally set as 1 or 2 s, the simulation time step is chosen as 0.2 s.
- *The sampling time to record flight data*. The sampling time of recording the flight attitude data has a great influence on the calibration accuracy. The larger this time, the larger the estimation errors. Considering the speed and capability of the on-board computer of UoSAT-12, it was chosen as 2 s in both the simulation and in-orbit experiments. Simulation results showed that with this sampling time, the estimation accuracy is still acceptable.
- *Errors in the satellite MOI*. The errors in the MOI tensor have very limited influence on the estimation accuracy of the thruster parameters. The influence occurs mainly in the transient response. In steady state, it is very small. These errors can therefore be neglected in the practical in-orbit tests.

Table 1
UoSAT-12 Simulation parameters

Orbit parameters	Orbital period 6000 s Inclination $i = 65^\circ$ Eccentricity $e = 0.0$
Satellite MOI tensor	$\mathbf{I} = \begin{bmatrix} 40.45 & -0.2 & -0.5 \\ -0.2 & 42.09 & 0.4 \\ -0.5 & 0.4 & 41.36 \end{bmatrix}$ kgm^2
Simulation time step	0.2 s
Sampling time of flight data	2 s
Quaternion feedback controller	5% Settling time = 180 s
Bang–Bang controller gains	$k_1 = 1.0$ (proportional gain) $k_2 = 2.3$ (derivative gain) $e_{band} = 0.02$ rad (dead band)
Reaction wheel	Z-axis wheel failed Maximum torque output = 0.015 Nm Moment of inertia of Y-wheel = 0.0077 kg m^2 Moment of inertia of X-wheel = 0.008 kg m^2 Maximum speed = 5000 rpm

- *Attitude measurement errors*. By adding different levels of white noise to the attitude measurements (roll, pitch and yaw angles), the effect on the estimation accuracy was investigated. For thruster parameters with larger values, the effect was quite small, but for those parameters with very small values, the estimation errors caused by measurement noise become much bigger, because the signal-to-noise ratio is very small. To improve the estimation accuracy subject to measurement noise, the following low-pass filter was proposed for the in-orbit calibration tests,

$$\hat{\mathbf{RPY}}(i) = \frac{\mathbf{RPY}(i-1) + 2\mathbf{RPY}(i) + \mathbf{RPY}(i+1)}{4}. \quad (26)$$

With the controller and estimation parameters derived during the simulation studies, the in-orbit experiments were also designed based on the simulation test results. The criteria for designing the firing time period for each specific thrusters were: (1) the maximum attitude changes should be within 5° in pitch and roll so as to guarantee accurate attitude measurements from the horizon and sun sensors, (2) the maximum torque required from the reaction wheels to stabilize the satellite must be within their allowable output levels. Simulation tests were conducted for all the ten cold-gas thrusters on-board UoSAT-12, as illustrated in Fig. 4. The appropriate firing times (from 1 to 3 s) were designed for each thruster, details can be found in Steyn et al. (2001).

Fig. 5 illustrates the simulation results for the thruster $\Delta V+$(Orbit + in Fig. 4), which produces mainly a negative pitch torque. During the simulation, the thruster was commanded to fire for 3 s starting at time

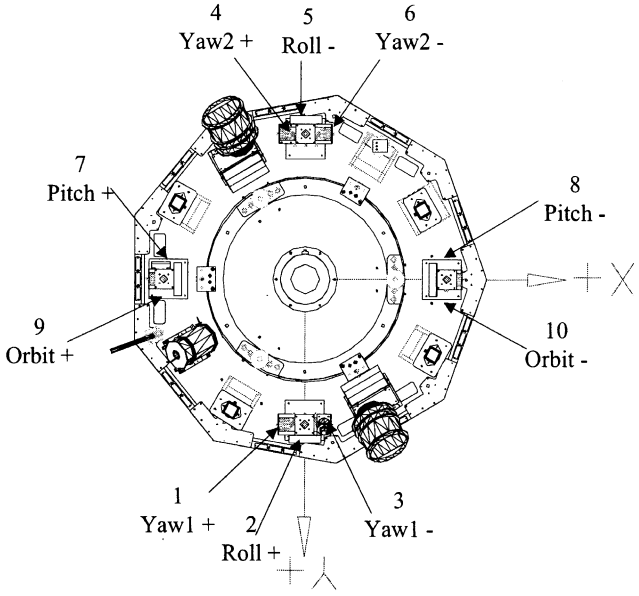


Fig. 4. Top view of cold gas thrusters on UoSAT-12.

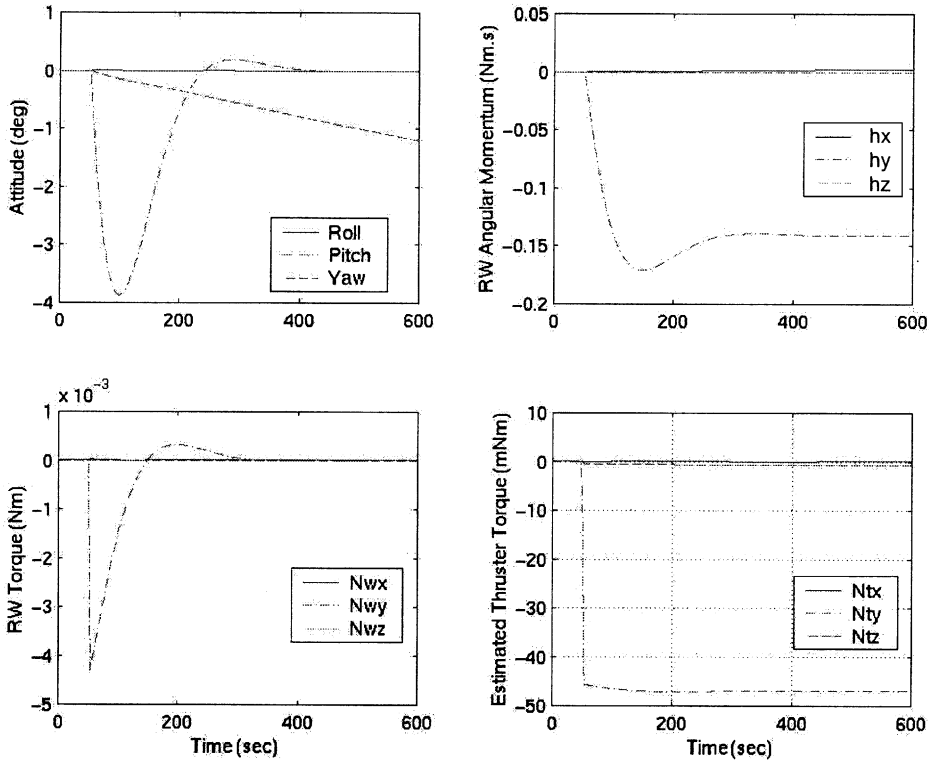


Fig 5. Simulation results of firing the thruster $\Delta V+$ for 3 s.

50 s. As can be seen, the maximum attitude disturbance in pitch is –3.87°, the maximum reaction wheel torque is –4.31 mNm, which are all within the design criteria. The modelled thruster parameters are [0, –47.0, –0.6] mNm, and the estimation results are [−0.0006, –46.9493, –0.5877] mNm, which shows a satisfactory estimation accuracy.

For the autonomous calibration method, the RLS algorithm was implemented together with a satellite simulator developed in the Matlab/Simulink environment (Bordany, 2001). During the simulation a set of three pairs of orthogonal thrusters were used to test the thruster algorithm to identify the $\mathbf{3}\times\mathbf{3}$ calibration matrix $\hat{\mathbf{A}}$. The RLS tuning parameters used in the simulation are listed in Table 2. Extensive simulation tests were conducted to evaluate the performances of the autonomous calibration approach. Two calibration algorithms, the RLS and a least-mean squares (LMS) method, and two disturbance generators, the Bang–Bang controller of Eq. (16) and an open-loop pseudo-random binary signal (PRBS) were used and compared to find the best calibration accuracy. Detailed simulation results and comparison analyses can be found in Steyn et al. (2001). Simulation results delivered the best performance with a Bang–Bang feedback controller and a RLS estimation algorithm. Fig. 6 illustrates the simulation results when estimating the thruster coefficient matrix. As can be seen, the thruster parameters converge to the true values in less than half an orbit.

6. In-orbit test results

Both calibration approaches were practically tested in-orbit on the minisatellite UoSAT-12. With the single thruster pulse method, in-orbit experiments were conducted for six thrusters, $\Delta V+$, $\Delta V-$, $P+$, $P-$, $R+$, and $R-$, during December 2001, see Steyn et al. (2001) for details. Each test firing starts from a three-axis stabilized nadir pointing attitude. Initially, the satellite is commanded to fly freely, without magnetorquer activity for 10 minutes, to record the external disturbances. Then one thruster is commanded to fire a short pulse (1–3 s) and the flight information are recorded for another 10 min, to estimate the thruster parameters. The recorded flight information were later downloaded to the ground station, and processed with the batch filter algorithms to estimate the corresponding thruster parameters.

To evaluate the consistency of the calibration results, multiple firing experiments (from 2 to 6 times) were conducted for each thruster, with the same or different firing time periods. Results showed a very good consistency for each thruster between different firing tests, which demonstrated the feasibility and practicability of the proposed calibration method. Fig. 7 illustrates the test results of the $\Delta V+$ thruster with a 3 s firing period. Compared to Fig. 5, it can be seen that the practical test results are quite similar to the simulation results. However, the test results are more

Table 2
RLS tuning parameters

RLS tuning parameters	Off-diagonal	Diagonal
X-axis	$\lambda = 0.997$ $P(0) = 10^4$ $a_{12}(0) = a_{13}(0) = 0.0$	$\lambda = 0.997$ $P(0) = 10^5$ $a_{11}(0) = 1.0$
Y-axis	$\lambda = 0.997$ $P(0) = 10^4$ $a_{21}(0) = a_{23}(0) = 0.0$	$\lambda = 0.997$ $P(0) = 10^5$ $a_{22}(0) = 1.0$
Z-axis	$\lambda = 0.997$ $P(0) = 10^4$ $a_{31}(0) = a_{32}(0) = 0.0$	$\lambda = 0.997$ $P(0) = 10^5$ $a_{33}(0) = 1.0$

noisy. The sharp pulses appearing in the estimated torque were caused by a discontinuity in the attitude measurements, caused by a temporary lost of horizon sensor measurements. As the pitch angle exceeded a certain value, the horizon sensor lost its measurements and the sun sensor and magnetometer (with a lower accuracy) become the only attitude measurement. However, in the steady state, the estimated thruster torque converges to a constant value, with a good consistency compared to other test results. Thus this kind of disturbances does not have great influences on the final estimation results. However, to prevent this kind of disturbance, the firing time was further reduced in the remaining tests to limit the attitude changes to be within the horizon sensor's FOV, and the discontinuities in the attitude measurements disappeared. Table 3 summarises the in-orbit calibration results for the $\Delta V+$, and $\Delta V-$ thrusters, which shows a good consistency between different firing tests.

Using the autonomous method, two thrusters, $\Delta V+$ and $\Delta V-$, were calibrated through in-orbit experiments. The pitch axis is disturbed using a Bang–Bang Y-reaction wheel controller ($N_{\max} = 2\,\mathrm{mN\,m}$), while the Y-axis is simultaneously controlled using the pair of ΔV thrusters to be calibrated. The X-axis is controlled using the X-wheel, while the Z-axis is controlled using the already calibrated yaw thrusters. The thrusters induced torque parameters are represented by the a 3×2 coefficient matrix, defined as

$$\mathbf{N}_T = \begin{bmatrix} a_{x\Delta vp} & a_{x\Delta vn} \\ a_{y\Delta vp} & a_{y\Delta vn} \\ a_{z\Delta vp} & a_{z\Delta vn} \end{bmatrix} \begin{bmatrix} T_{\Delta vp} \\ T_{\Delta vn} \end{bmatrix}, \tag{27}$$

where $\mathbf{N}_T$ is the torque vector induced by the two thrusters, $T_{\Delta vp}$ is the maximum expected torque level (negative) of the $\Delta V+$ thruster, and $T_{\Delta vn}$ is the maximum expected torque level (positive) of the

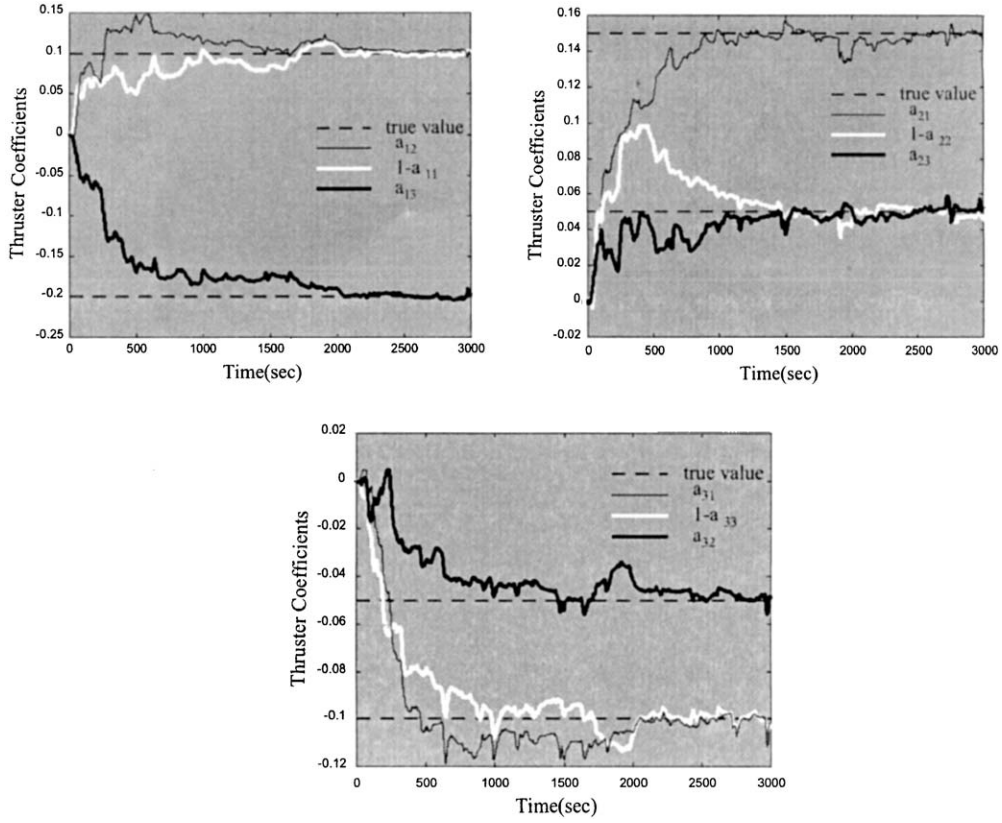


Fig. 6. Estimated thruster coefficients without aerodynamic disturbance.

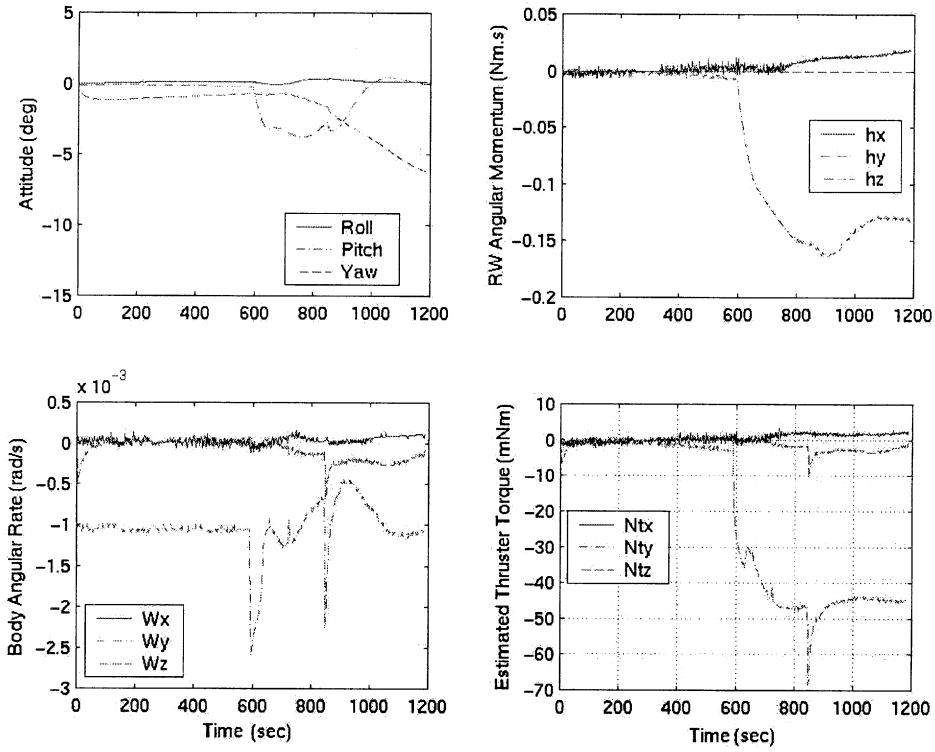


Fig. 7. $\Delta V+$ thruster firing (3 s) results with the low-pass filtering.

Table 3
In-Orbit Torque Estimation Results (in body axis, mN m)

Thruster	Test 1	Test 2	Test 3	Average
$\Delta V+$	0.6395	−0.5477		0.0729
	−39.1316	−39.1760	—	−39.1538
	−3.1762	−3.2430		−3.2096
$\Delta V-$	1.7584	0.9628	0.5851	1.1021
	38.0184	37.5720	37.3414	37.6439
	2.5469	4.4942	2.1508	3.0640

ΔV-thruster, and $\mathbf{A}_c$ is the 3×2 thruster coefficient matrix to be identified. Based on pre-launch ground test results, the maximum expected torque level for both ΔV thrusters is set as 46.5 mNm.

The parameters used for the experiment are the same as in Table 1. Fig. 8a illustrates the wheel momentum during the experiment. The drift in the wheel momentum for the Y-axis is due to the cross disturbances caused by the Z-thrusters. As is clear from Fig. 8b the pitch angle does not exceed $\pm 3^\circ$. The angular rate in the Y-axis is not triangular in shape (Fig. 8c), this is due to the feedback ΔV thruster controller, when the Y-axis is disturbed with the Y-wheel. Fig. 8d shows the thruster torque during the experiment. The estimated thruster coefficients using the RLS estimation method are shown in Figs. 8e and f. The mean values of the thruster coefficients converge approximately to the following values:

$$\begin{bmatrix} a_{x\Delta vp} & a_{x\Delta vn} \\ a_{y\Delta vp} & a_{y\Delta vn} \\ a_{z\Delta vp} & a_{z\Delta vn} \end{bmatrix} = \begin{bmatrix} 0.012 & -0.014 \\ 0.829 & 0.805 \\ 0.01 & -0.005 \end{bmatrix}. \quad (28)$$

The total in-orbit calibration results with the two approaches are summarised in Table 4. Details for each thruster firing can be found in Steyn et al. (2001). As can be seen, the estimation results converge to almost the same dominant values for the ΔV thrusters, which demonstrates a good consistence in the two proposed calibration approaches. But for the non-dominant values along the undisturbed axes, e.g. the values around x and z axes for the thrusters $\Delta V+$ and $\Delta V-$, the single thrusters pulse method gives larger errors as compared to the autonomous method. This can be explained by the fact that the time variation in disturbance torques was not very well compensated in the single thrusters pulse method. By firing a thrust pulse, the satellite attitude changes, though in a very small scale, thus the external disturbance torques do not apply in the same way as in the first stage when recording the external disturbances. Meanwhile the real value around undisturbed axes are around the level of zero, thus this time-variant external disturbance deteriorates the accuracy in the calibration results.

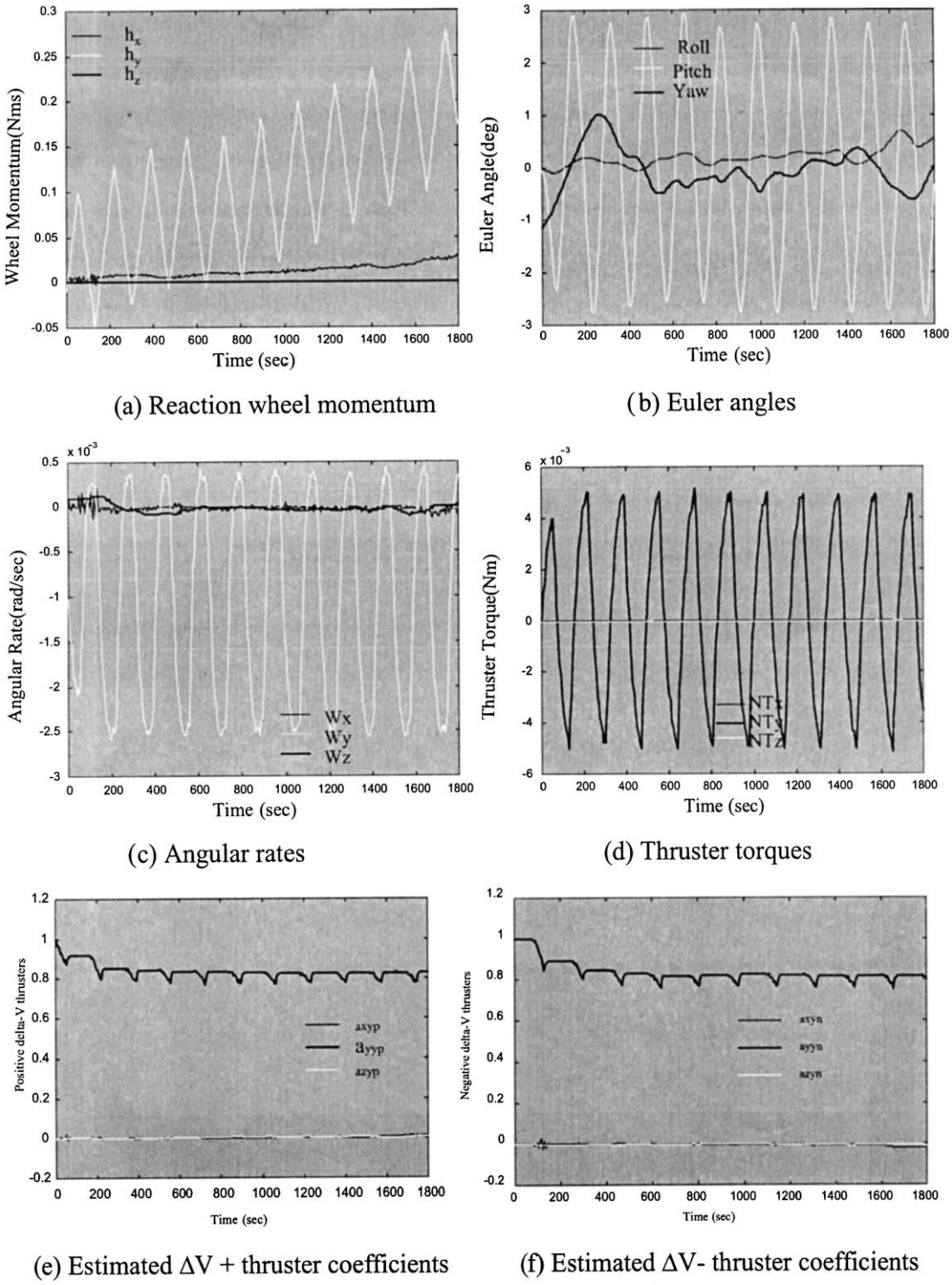


Fig. 8. UoSAT-12 responses during the ΔV thrusters calibration. (a) Reaction wheel momentum; (b) Euler angles; (c) Angular rates, (d) Thruster torques, (e) Estimated $\Delta V+$ thruster coefficients and (f) Estimated $\Delta V-$ thruster coefficients.

When compared to their model values, which were calculated based on the pre-launch ground test results, as given in Table 4, all the torque values are found to be at a lower level. This means that the model values were miscalculated, or the ground test results were not very accurate and reliable. It also demonstrates the importance of the in-orbit real-time calibration for thrusters on-board a satellite in practical engineering.

Table 4
Summary of the in-orbit test results for thrusters on-board UoSAT-12

$\vec{N}_{Ti} = \begin{bmatrix} N_{xi} \\ N_{yi} \\ N_{zi} \end{bmatrix}$ (mN m)	$\Delta V+$	$\Delta V-$	$P-$	$P+$	$R+$	$R-$
Model value	0	0	0.57	0.57	66.42	−65.28
	−46.95	46.95	−66.975	64.725	−1.125	−1.125
	−0.57	0.57	0	0	0	0
Autonomous method	−0.558	−0.651				
	−38.549	37.957	—	—	—	—
	−0.465	−0.233				
Single thruster pulse method	0.0729	1.1021	−1.9541	0.3473	60.3149	−58.721
	−39.154	37.6439	−60.677	57.4945	−3.0140	−5.7324
	−3.2096	3.0640	−1.0613	1.2674	0.3054	0.2884

7. Conclusion

Two approaches for in-orbit thruster calibration, a single thruster pulse method and an autonomous method, are presented, studied through simulation analyses and evaluated through in-orbit implementation and firing tests.

In the single thruster pulse method, one thruster is commanded to fire for a short time to produce a pulse external torque disturbance, then the transient response of satellite attitude and the internal reaction wheel control system are recorded. A batch filter algorithm is then used to process the recorded flight data to provide an estimation of the firing thruster's parameters.

In the autonomous method, a repetitive torque disturbance (by using the on-board reaction wheels) is applied to the satellite, the cold-gas thrusters are then applied in a feedback control system to stabilize the satellite attitude. By measuring the resulted attitude disturbances and the thruster controller outputs, a recursive least square or least mean square algorithm is used to estimate the thruster parameters.

Simulation analyses were conducted for both methods to evaluate their feasibility and performance, to evaluate the in-orbit calibration experiments, and to verify the performance of the developed filtering algorithms.

In-orbit experiments were implemented on-board UoSAT-12. Satisfactory estimation results were obtained, which reveal that both the approaches work well with an acceptable accuracy. For the dominant torque values, both methods give a very convergent result, while for non-dominant torque values, the autonomous method would give a better accuracy.

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